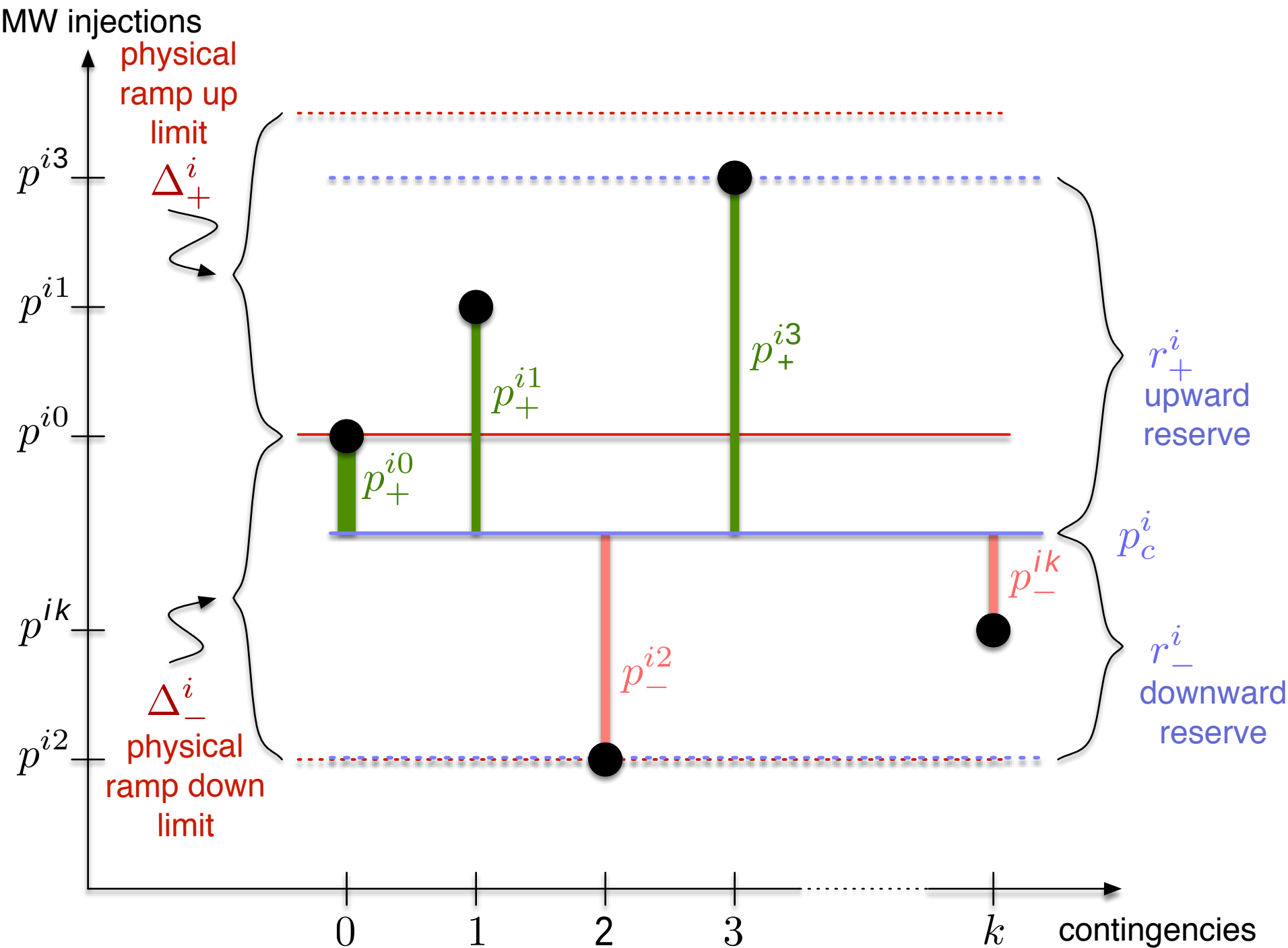
MW injections
physical ramp up limit
Δ_+^i
p^{i3}
p^{i1}
p^{i0}
p^{ik}
p^{i2}
Δ_-^i
physical ramp down limit
p_+^{i1}
p_+^{i3}
p_+^{i0}
p_-^{ik}
p_-^{i2}
r_+^i
upward reserve
p_c^i
r_-^i
downward reserve
0
1
2
3
k
contingencies